Updating parameters

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

TEACHER GUIDE · 1 OF 9

Prepare the purpose, prerequisites and materials

> Prepare Updating parameters: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Apply a **learning rate** to an ordered gradient

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Gradients with a numerical check

Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

Look for: Estimate one partial derivative while other parameters stay fixed

Reference example, not the only acceptable wording: Teacher explanation: **Prediction** 3, residual 0, loss 0, gradient [0, 0].

Loss functions

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate per-example squared error and mean squared error

Reference example, not the only acceptable wording: Teacher explanation: Residuals are -1 and 2, squares 1 and 4, MSE 2.5.

Composing layers

Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

Look for: Trace how one layer supplies the next layer **input**s

Reference example, not the only acceptable wording: Teacher explanation: Scores are -2 and 2; activations are 0 and 2; y = 0 - 2 + 1 = -1.

For the compact alternative schedule, use two 45-minute sessions and keep old and new parameter boxes separate. This lesson checks one update, not a full optimiser.

Treat the mixed-state procedure as different from the stated algorithm, rather than claiming all sequential methods are invalid.

Outcomes

Apply a **learning rate** to an ordered gradient

Update all parameters from the same saved state

Verify the new loss and identify **overshoot**ing

Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Prerequisite answer

The residual is -1 and the gradient is [-4, -2].

End goal for the pictorial case

By the end, I can explain updating parameters using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain and model the mechanism

> Model Updating parameters using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Gradient descent uses new parameter = old parameter - **learning rate** times its gradient entry. The minus sign chooses the locally downhill direction for a positive **learning rate**.

A negative gradient therefore increases the corresponding parameter.

Keep the Week 10 example: x = 2, target 3, y = 2w + b and L = (y - 3) squared. At [w, b] = [1, 0], the loss is 1 and gradient is [-4, -2].

With **learning rate** 0.1, new w = 1 - 0.1 times -4 = 1.4 and new b = 0 - 0.1 times -2 = 0.2.

This particular step reaches zero loss on one example. It does not establish a unique parameter solution, successful learning on other examples or a generally ideal **learning rate**.

Explain Updating parameters · 1

Predict the rows: unchanged parameters and loss 1 at rate 0; [1.2, 0.1] and loss 0.25 at rate 0.05; [1.4, 0.2] and loss 0 at rate 0.1; [2.2, 0.6] and loss 4 at rate 0.3.

The helper performs one step on this fixed example; it assumes numeric **input**s and does not choose a **learning rate** for you.

Term	Meaning
Learning rate	A multiplier that controls the size of a gradient update
Gradient descent	Updating parameters in the direction opposite the gradient
Simultaneous update	Computing all new parameter values from the same old state
Overshoot	A step that moves too far and can increase the objective

Use one gradient for one parameter snapshot

STAGE	SAVED VALUES	ACTION
Before the step	w is 1 and b is 0	Compute gradient [-4, -2]
Apply rate 0.1	Use both old parameters	Produce [1.4, 0.2]
After the step	Use both new parameters	Recompute loss 0

Read top to bottom. All gradient entries come from the same old model. The final row evaluates the fully updated model rather than mixing old and new coordinates.

Guided: [1.2, 0.1] gives **prediction** 2.5, residual -0.5 and loss 0.25. Exit: no.

The mixed-state procedure lowers this loss but does not implement the specified simultaneous gradient step; compare states and gradients, not just whether loss decreased.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Use one gradient for one parameter snapshot

STAGE	SAVED VALUES	ACTION
Before the step	w is 1 and b is 0	Compute gradient [-4, -2]
Apply rate 0.1	Use both old parameters	Produce [1.4, 0.2]
After the step	Use both new parameters	Recompute loss 0

Read top to bottom. All gradient entries come from the same old model. The final row evaluates the fully updated model rather than mixing old and new coordinates.

Starting again from [1, 0], use learning rate 0.05. Show both new parameters, the new prediction and the new loss.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Updating parameters

Use one explicit gradient update and then recompute the loss.

The facts we will use

Supplied facts

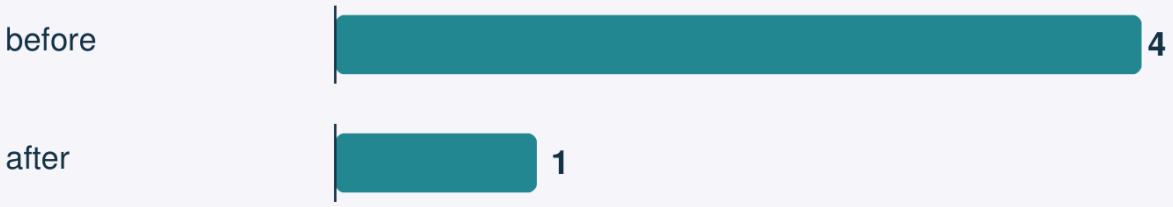
Model y=2w+b, target 5. Start w=1, b=1; **learning rate** 0.05. Initial y=3, error y−target=−2, squared loss 4.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/11

Loss before and after rate 0.05 update



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/11

Updating parameters

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute gradient

Gradient for [w, b]=[2 x (-2) x 2, 2 x (-2)]=[-8,-4].



2. Update simultaneously

New w=1-0.05 x (-8)=1.4; new b=1.2.



3. Check new loss

New y=4, residual -1, loss 1: lower than 4.

CHANGE ONE THING

Set learning rate 0.2 from the same starting point.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute gradient	Gradient for [w, b]=[2×(-2)×2, 2×(-2)]=[-8,-4].
2. Update simultaneously	New w=1-0.05×(-8)=1.4; new b=1.2.
3. Check new loss	New y=4, residual -1, loss 1: lower than 4.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Gradient descent subtracts **learning rate** times gradient. A negative gradient makes this subtraction increase the parameter. Compute both gradients at the original [w,b] before updating either one.

The small step brings the **prediction** closer to five, reducing squared error. A larger step can cross the target and undo that gain, so the new **prediction** and loss must be recalculated.

What this example does not establish

Keep the stated limit

One successful toy update is not **evidence** of convergence for a real network or generalisation to unseen data.

Change one thing in the pictured case

Change one thing

Change: Set **learning rate** 0.2 from the same starting point.

Predicted result

Prediction: New w=2.6, b=1.8, y=7, squared loss 4: no improvement.

Why the result follows

Reason: The larger update jumps to an equally distant point on the opposite side of the target.

Ask while showing the picture

Ask while showing the picture · 2

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Updating parameters, reduce support deliberately and check individual reasoning before discussing answers.

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the Week 10 gradient and write the update rule with parentheses around negative entries.
7 to 18 minutes	Calculate the rate 0.1 worked step and recompute the loss from the new parameters.
18 to 32 minutes	Complete W1 and the guided rate 0.05 calculation.
32 to 45 minutes	Run or trace all four code rows and complete W2.
Session 2 start to 8 minutes	Rebuild the old-state snapshot and compare it with the mixed-state example.
8 to 20 minutes	Complete W3 and discuss zero gradient with W5.
20 to 35 minutes	Collect W4 independently, including old loss, both new parameters and new loss.
35 to 45 minutes	Use the exit response and a fresh retry. Keep the model-quality claim separate from implementation correctness.

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Use separate old, gradient and new columns. Extension: for the lesson example, show that the new residual is the old residual multiplied by (1 - 10 times rate).

Explain why rates 0 and 0.2 preserve loss while 0.3 raises it for a nonzero residual.

Reduce help deliberately

Completion practice

Completion task: For the lesson start [1, 0], apply rate 0.3 and calculate **prediction** and loss.

Reasoning to listen for

Teacher check: Teacher explanation: Gradient [-4, -2] gives [2.2, 0.6], **prediction** 5 and loss 4, larger than the starting loss 1.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Updating parameters. Proposed explanation: “Subtracting a negative gradient decreases the parameter”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> ### Expected repair
>
> Expected correction: Subtracting a negative adds a positive amount; show the signed calculation.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

A gradient supplies a local direction, but step size can **overshoot**. Rate 0.2 gives the same loss 4 despite using the correct gradient. Changed-case check: New w = 2.6, b=1.8, y = 7, loss 4: no improvement.

The larger update jumps to an equally distant point on the opposite side of the target.

Changed-case reasoning: New w=2.6, b=1.8, y=7, squared loss 4: no improvement. The larger update jumps to an equally distant point on the opposite side of the target.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the investigation and diagnose misconceptions

> Run the Updating parameters investigation and use the first unsupported step to choose a teaching response.

Subtracting a negative gradient decreases the parameter: Subtracting a negative adds a positive amount; show the signed calculation.

A downhill direction allows any rate: A finite step can **overshoot** and increase loss.

Recompute each gradient after its assignment: For this algorithm, all entries are computed at the same saved state.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Updating parameters questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

Guided: [1.2, 0.1] gives **prediction** 2.5, residual -0.5 and loss 0.25. Exit: no.

The mixed-state procedure lowers this loss but does not implement the specified simultaneous gradient step; compare states and gradients, not just whether loss decreased.

W1 Apply a negative gradient

Workbook question · W1

Starting with parameter a = 2, gradient -3 and **learning rate** 0.2, calculate the new a.

Teacher reasoning · W1

Teacher explanation: New a = 2 - 0.2 times -3 = 2.6.

W2 Detect overshooting

Workbook question · W2

For the lesson start [1, 0], apply rate 0.3 and calculate **prediction** and loss.

Teacher reasoning · W2

Teacher explanation: Gradient [-4, -2] gives [2.2, 0.6], **prediction** 5 and loss 4, larger than the starting loss 1.

W3 Preserve one snapshot

Workbook question · W3

Explain why recomputing the bias gradient after changing w is not the **simultaneous update** specified here.

Teacher reasoning · W3

Teacher explanation: The two gradient entries then belong to different parameter vectors. Save the full old gradient and use it for both assignments.

W4 Update a new model independently

Workbook question · W4

Use x = 1, target 4, w = 1, b = 1, squared loss and **learning rate** 0.1. Compute **prediction**, residual, gradient [2 times residual times x, 2 times residual], new parameters and new loss.

Teacher reasoning · W4

Teacher explanation: **Prediction** 2, residual -2, gradient [-4, -4]. New parameters [1.4, 1.4] give **prediction** 2.8, residual -1.2 and loss 1.44, down from 4.

W5 Evaluate a zero gradient

Workbook question · W5

At a point with gradient [0, 0], what does this update do? Does that alone prove a global minimum on any loss surface?

Teacher reasoning · W5

Teacher explanation: It leaves parameters unchanged for a finite **learning rate**. Zero gradient alone does not establish a global minimum; stationary points can have other meanings.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4 requires gradient [-4, -4], parameters [1.4, 1.4] and loss 1.44. Fresh retry: x = 1, target 1, w = 1, b = 1, rate 0.25 gives residual 1, gradient [2, 2], new parameters [0.5, 0.5] and loss 0.

Require both updates to use the original residual.

Assess independent transfer

Independent question · withhold the key

Selected case: Use x = 1, target 4, w = 1, b = 1, squared loss and **learning rate** 0.1. Compute **prediction**, residual, gradient [2 times residual times x, 2 times residual], new parameters

and new loss.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: **Prediction** 2, residual -2, gradient [-4, -4]. New parameters [1.4, 1.4] give **prediction** 2.8, residual -1.2 and loss 1.44, down from 4.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect the project, support and next lesson

> Connect Updating parameters to Image-pattern Network Investigation, adapt support and plan a later changed-case check.

Add a labelled updating parameters **evidence** sheet to Image-pattern Network Investigation. Show how you can apply a **learning rate** to an ordered gradient.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Use separate old, gradient and new columns. Extension: for the lesson example, show that the new residual is the old residual multiplied by (1 - 10 times rate).

Explain why rates 0 and 0.2 preserve loss while 0.3 raises it for a nonzero residual.

Week 12 repeats the **training** cycle for a small network.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/12, ai-5/14, ai-5/17. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	

Record	Teacher evidence
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

TEACHER GUIDE · 7 OF 9

Prepare, teach and assess

> Use this preparation summary alongside the detailed Updating parameters lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Apply a **learning rate** to an ordered gradient

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: For the lesson start [1, 0], apply rate 0.3 and calculate **prediction** and loss.

> **Supported practice answer**
>
> Reasoned practice answer: Teacher explanation: Gradient [-4, -2] gives [2.2, 0.6], **prediction** 5 and loss 4, larger than the starting loss 1.

Check every learner before discussion: Use the worked case for Updating parameters. Proposed explanation: “Subtracting a negative gradient decreases the parameter”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Subtracting a negative adds a positive amount; show the signed calculation.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Set **learning rate** 0.2 from the same starting point. Require a **prediction** and a reason before comparing. Expected result: New w=2.6, b=1.8, y=7, squared loss 4: no improvement.

Reason: The larger update jumps to an equally distant point on the opposite side of the target.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately

Criterion	Not yet	With support	Independent
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

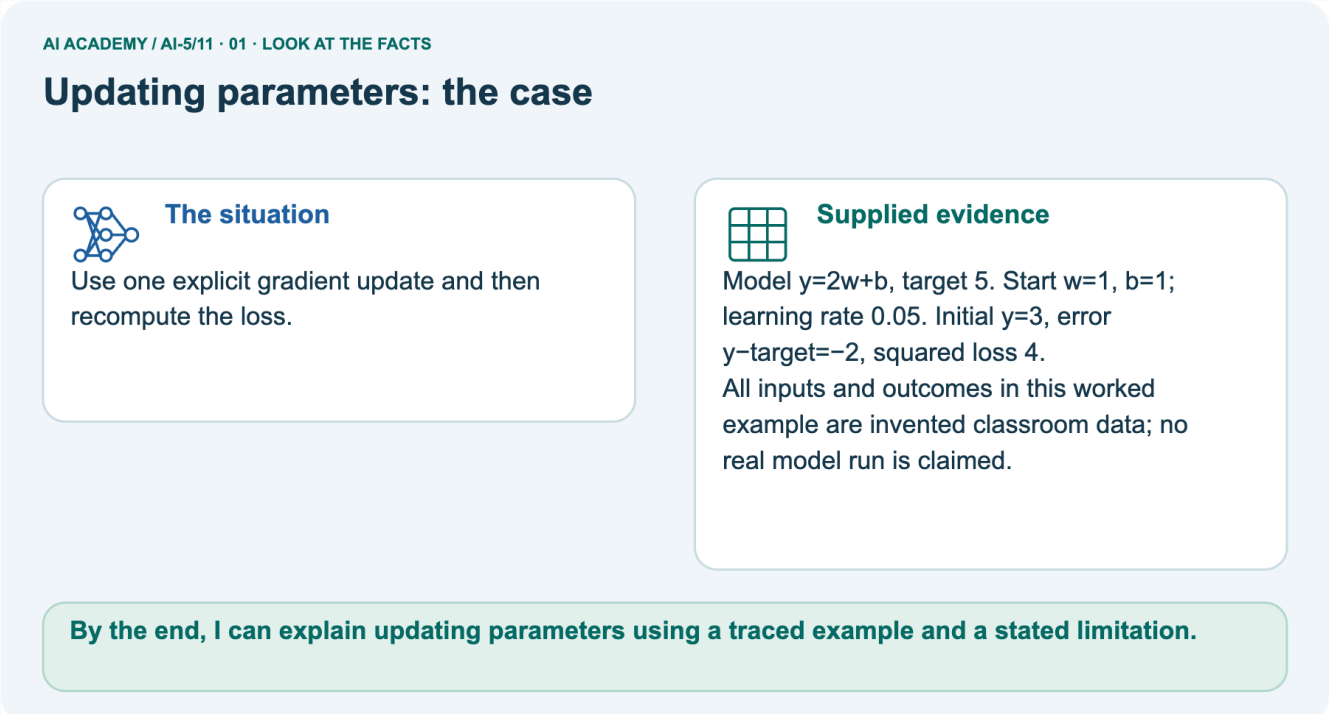
TEACHER GUIDE · 9 OF 9

Visual concept journey

> Show one Updating parameters graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Updating parameters: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Use one explicit gradient update and then recompute the loss. Supplied facts: Model y=2w+b, target 5. Start w=1, b=1; **learning rate** 0.05. Initial y=3, error y−target=−2, squared loss 4.

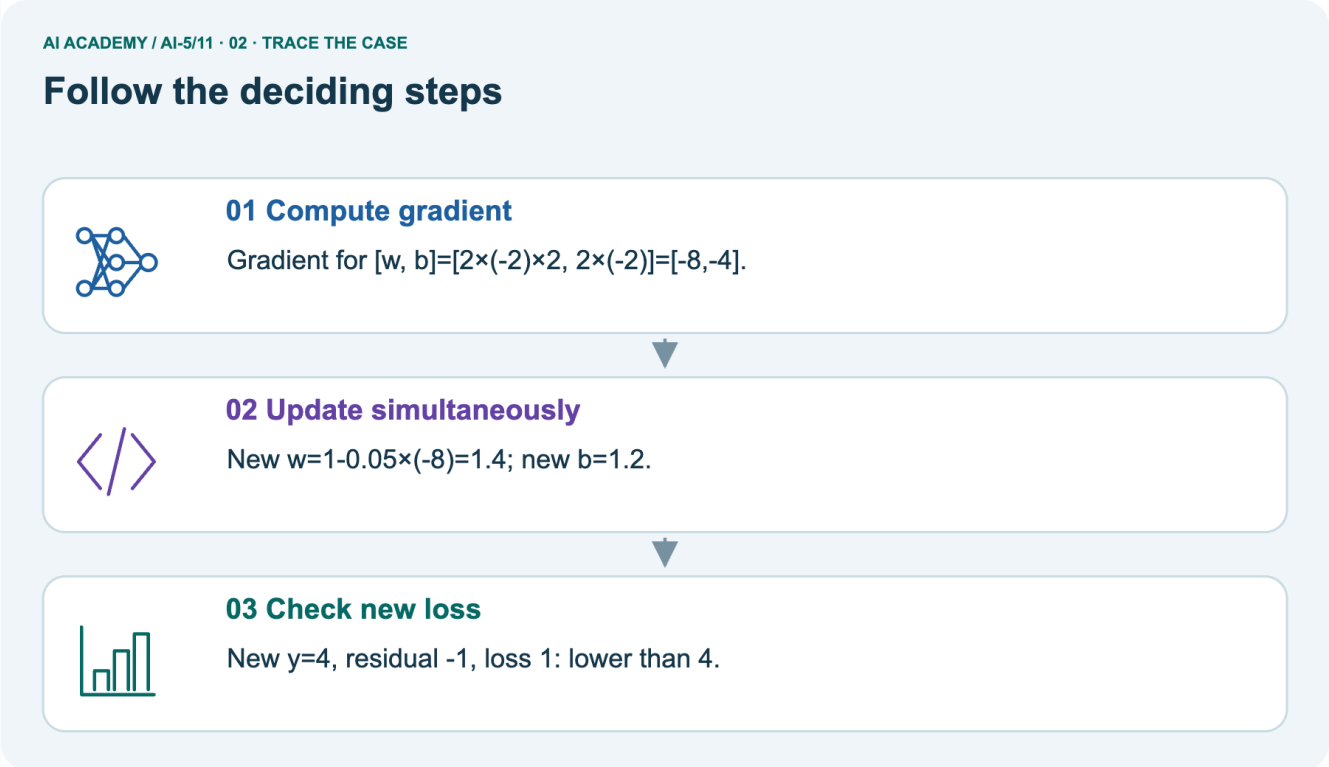
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

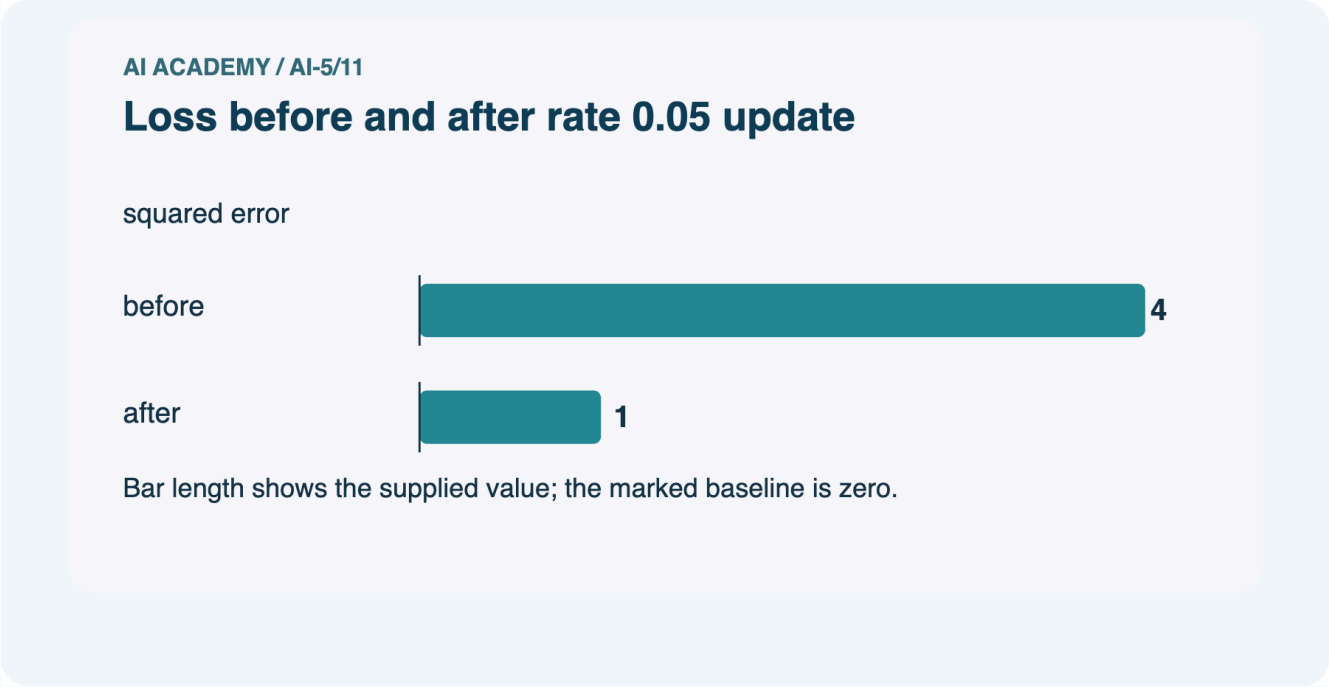
Compute gradient: Gradient for [w, b]=[2×(-2)×2, 2×(-2)]=[-8,-4]. → Update simultaneously: New w=1-0.05×(-8)=1.4; new b=1.2. → Check new loss: New y=4, residual -1, loss 1: lower than 4.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Loss before and after rate 0.05 update



Follow the labelled picture with the matching explanation. Open the vector diagram

Loss before and after rate 0.05 update. Units: squared error. before: 4. after: 1.

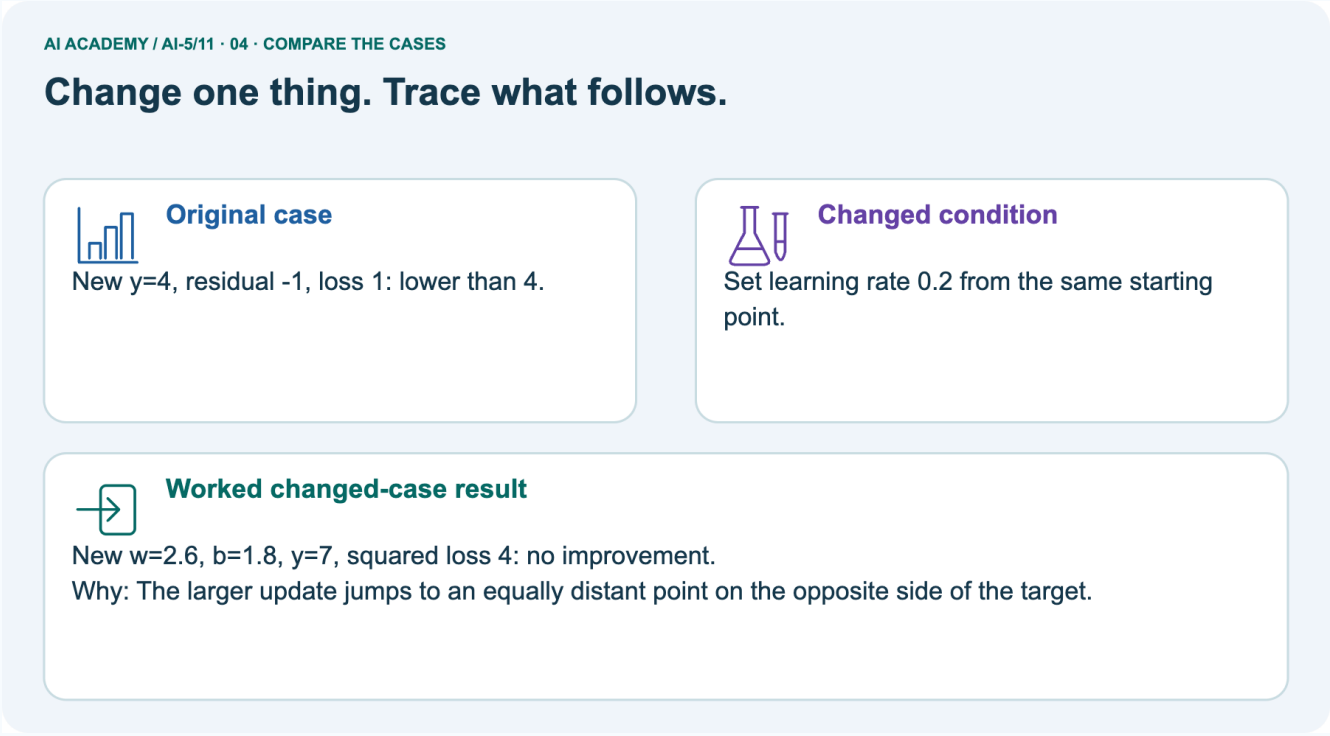
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: New y=4, residual -1, loss 1: lower than 4. Changed condition: Set **learning rate** 0.2 from the same starting point. Expected result: New w=2.6, b=1.8, y=7, squared loss 4: no improvement.

Reason: The larger update jumps to an equally distant point on the opposite side of the target.

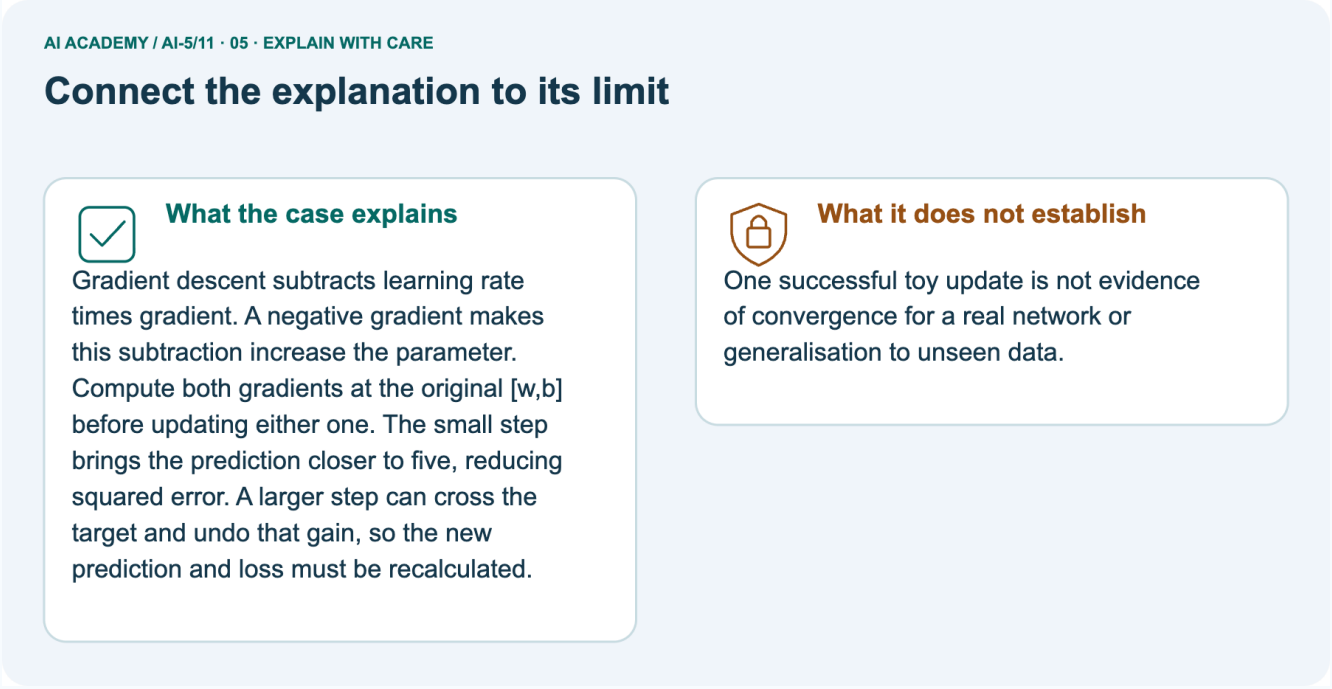
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The larger update jumps to an equally distant point on the opposite side of the target.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Gradient descent subtracts **learning rate** times gradient. A negative gradient makes this subtraction increase the parameter. Compute both gradients at the original [w,b] before updating either one.

The small step brings the **prediction** closer to five, reducing squared error. A larger step can cross the target and undo that gain, so the new **prediction** and loss must be recalculated.

Limit: One successful toy update is not **evidence** of convergence for a real network or generalisation to unseen data.

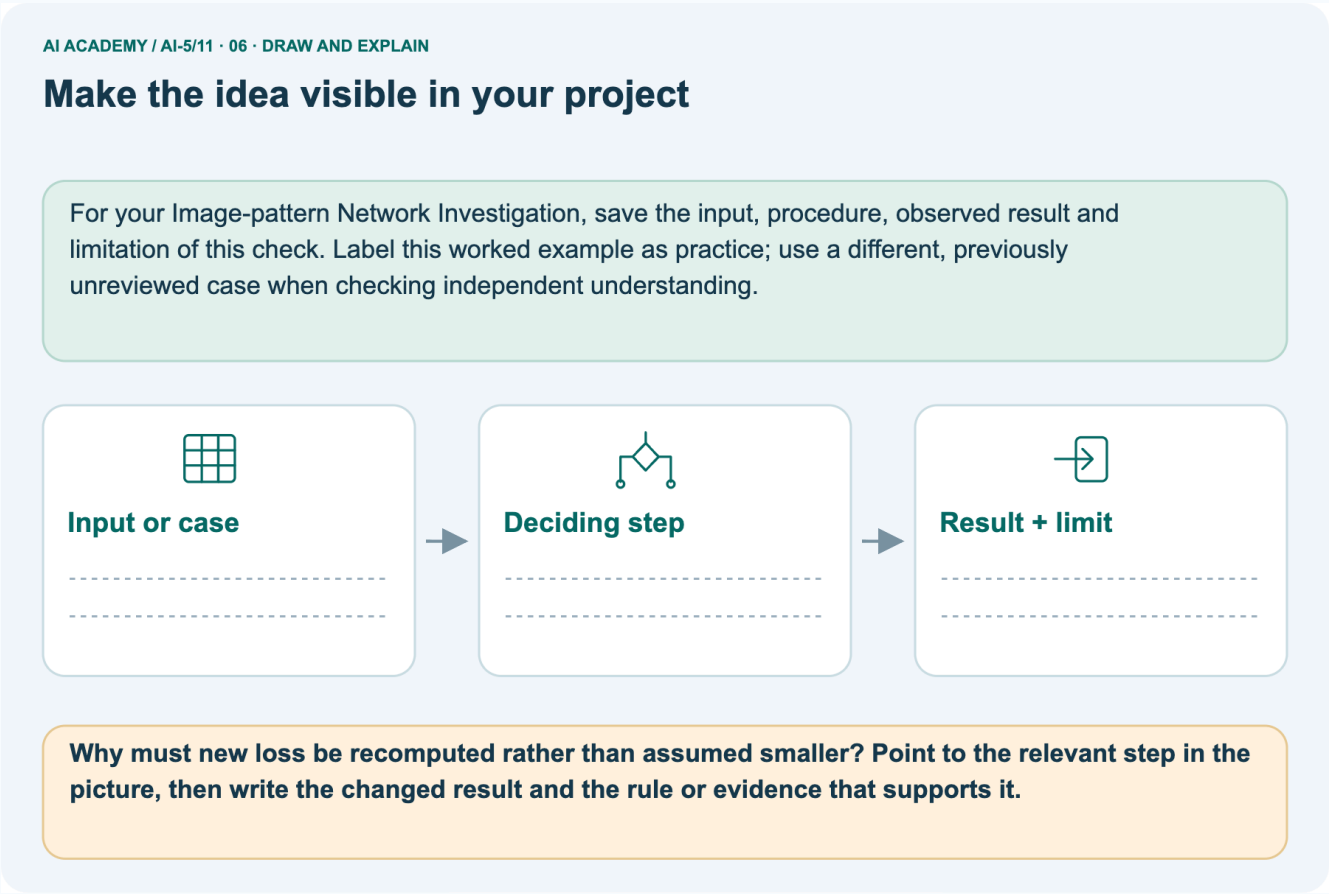
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: One successful toy update is not **evidence** of convergence for a real network or generalisation to unseen data.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: A gradient supplies a local direction, but step size can **overshoot**. Rate 0.2 gives the same loss 4 despite using the correct gradient.

Changed-case check: New w = 2.6, b=1.8, y = 7, loss 4: no improvement. The larger update jumps to an equally distant point on the opposite side of the target.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.